

The Case for Voluntary Disclosure: Strategic Feature Revelation in Regression and Classification

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Abstract

This paper addresses a fundamental design question: should a predictive model enforce mandatory feature disclosure or allow voluntary disclosure by participants? Since participants strategically reveal only those features that align with their private objectives, the model must account for opportunistic reporting. For the regression problem, we establish conditions under which the predictive model actually achieves better performance by permitting selective revelation. We extend these results to the classification setting and provide specialized insights for Gaussian distributions. Finally, we explore a dynamic learning framework where the model interacts repeatedly with participants, presenting an algorithm that adaptively switches between disclosure regimes with established theoretical performance bounds.

1 INTRODUCTION

The performance of any predictive model depends on the set of features provided for prediction. In standard settings, all relevant features are assumed to be fully observed by the model. In this paper, we study a setting in which a participant can voluntarily choose which features to disclose. It is natural to assume that the participant will disclose only those features that are beneficial to their personal objective. The central question we address is: under what conditions does voluntary feature disclosure actually improve predictive performance?

For a predictive model to benefit from such disclosure, its inference procedure must account for the possibility of opportunistic reporting. This problem becomes intricate when participants do not share a uniform directional preference over outcomes, as these heterogeneous preferences make it difficult to exploit systematic patterns in feature withholding.

A concrete and illustrative example arises in financial markets when estimating a firm's valuation. The market may enforce a mandatory disclosure rule for financial information. If full disclosure is required, the resulting valuation achieves a baseline accuracy dependent on the complexity of the underlying relationship. Alternatively, the market might allow firms to voluntarily disclose financial details, in which case the valuation is formed based only on the revealed information.

Firms typically hold privately preferred valuations that differ from the market's assessment, and the preferred direction of this deviation is not uniform across firms. Some firms actively seek overvaluation because it raises stock prices and lowers the cost of equity financing. Others may prefer undervaluation, for example, when preparing for a stock buyback program or seeking to avoid triggering regulatory scrutiny.

If a firm's preferred valuation is closely aligned with the market's objective, selective feature disclosure may assist by intuitively filtering out noisy or less informative signals, thereby enhancing valuation accuracy relative to mandatory disclosure. In contrast, if the firm's preferences are misaligned with the market's objective, selective disclosure distorts the available information and introduces a systematic bias, thereby degrading the valuation accuracy.

In the special case if firms shared the same directional preference—for instance, if every firm preferred overvaluation—the market could prevent exploitation simply by assigning a conservatively low default valuation in cases of partial disclosure. Such a rule would effectively discourage withholding, thereby replicating the outcome of a mandatory disclosure regime. In this work, however, we accommodate the more general and realistic setting in which participants hold heterogeneous preferences. We derive conditions to identify when the strategic disclosure of features is beneficial to the predictive model.

1.1 Main Contributions

In our work, we consider the predictive problems of classification and regression, modeling the strategic interaction as a sender-receiver game.

Firstly, assuming the environment is fully known to both players, we establish sufficient conditions in the regression setting for the predictive model to benefit from voluntary disclosure. These conditions depend on the misalignment between the sender’s preferred value and the ground truth value. We also provide necessary conditions that depend on the relationship between the disclosed features and the ground truth.

Secondly, we extend our analysis to the classification problem, establishing necessary and sufficient conditions for strategic advantage in terms of the structure of various conditional distributions.

Thirdly, we explore a learning version of the problem. There are two variants of the learning problem: static learning and online learning. In static learning, the receiver has access to a population sample of the environment and must select the disclosure regime and predictor to minimize the empirical risk. In online learning, we study a sequential strategic regression under partial feature observation, where the receiver learns environment parameters over time. We present online optimization algorithms using surrogate updates for this setup and establish theoretical convergence bounds to local optima.

1.2 Related Works

Our work builds on the intersection of strategic machine learning, sender-receiver games, and voluntary disclosure in predictive modeling. Early studies in strategic classification, such as [1], formalized settings where agents manipulate features to influence outcomes, showing that robust classifiers can mitigate gaming in linear models. This was extended in [2] to dark (unknown) environments, where agents’ responses are not fully observable, leading to equilibrium analyses under uncertainty.

Sender-receiver games provide a foundational framework for our analysis. [3] examined learning dynamics in these games using experimental data, highlighting how repeated interactions evolve into efficient signaling. Dynamic variants, like those in [4], incorporate Markov chains for state evolution, akin to our noisy channel model. Evolutionary perspectives, as in [5], explore multi-agent signaling, while [6] compared stimulus-response and belief-based learning, informing our online regret bounds.

Voluntary disclosure literature addresses selective revelation. [7] modeled evolving news in dynamic settings, deriving conditions for disclosure thresholds. [8] analyzed investor disclosures, predicting good-news bias, which parallels our heterogeneous preferences. Empirical studies, such as [9], used XGBoost and XAI for predicting disclosure patterns, emphasizing transparency in strategic contexts.

Stackelberg equilibria are central to our game-theoretic approach. [10] characterized equilibria in strategic classification and regression, designing incentives to prevent cheating. [11] generalized

PAC-learning for heterogeneous agents, bridging classification and regression. In adversarial prediction, [12] framed problems as Stackelberg games, solving for optimal decision rules.

Online learning with strategic agents extends our dynamic framework. [13] studied allocation under strategic behavior, deriving regret-minimizing algorithms. [14] proposed online principal-agent models for unknown types, achieving sublinear regret. Decentralized settings in [15] model coalition formation among selfish agents, aligning with our adaptive disclosure policy.

While these works advance strategic aspects in classification [16] and regression [17], they often assume known environments or linear models. Our contributions provide formal sufficiency and necessity conditions under noisy channels and heterogeneous preferences, with theoretical bounds for online adaptation, filling a gap in generalized predictive settings.

2 PROBLEM FORMULATION

Let $\mathcal{X} = \mathbb{R}^d$ be the feature space and $\mathcal{Y} = \mathbb{R}$ (or $\mathcal{Y} = \{-1, 1\}$ for classification) be the outcome space. We consider a data-generating process where Nature selects $(u, x, y) \sim (U, X, Y)$ over $\mathcal{Y} \times \mathcal{X} \times \mathcal{Y}$. Here, y is the ground truth value, x represents the available features, and u is the preferred value.

The problem is modeled as a game between two players: a **Sender** and a **Receiver**. The Sender observes (u, x) and chooses an action A from the strategy space:

$$\mathcal{S} = \{A : \mathcal{Y} \times \mathcal{X} \mapsto \{\emptyset, x\}\}.$$

This action represents the Sender's decision to either disclose the full feature vector x or withhold it (denoting the latter by $\emptyset$). The chosen action $A(u, x)$ is transmitted through a noisy channel, resulting in an observed message M . The channel noise is characterized by a parameter $q \in [0, 1]$ such that:

$$M = \begin{cases} \emptyset & \text{if } A(u, x) = \emptyset, \\ \emptyset & \text{if } A(u, x) = x \text{ with probability } q, \\ x & \text{if } A(u, x) = x \text{ with probability } 1 - q. \end{cases}$$

Upon observing $M \in \{\emptyset, x\}$, the Receiver applies a strategy $\hat{y} \in \mathcal{R}$:

$$\mathcal{R} = \{\hat{y} : \{\emptyset, x\} \mapsto \mathcal{Y}\}.$$

The loss of each player is defined as follows: the Sender incurs a loss $\ell(u, \hat{y}(M))$ based on their preferred value u , while the Receiver incurs a loss $\ell(y, \hat{y}(M))$ based on the ground truth value y .

We analyze this interaction as a **screening game**. In this game, the Receiver first commits to a strategy $\hat{y} \in \mathcal{R}$. Then, the Sender observes this commitment and takes an optimal action $A \in \mathcal{S}$ based on $\hat{y}$. Finally, the Receiver executes their committed strategy and the outcome is realized.

For any commitment $\hat{y}$ by the Receiver, the Sender's **best response** is to choose an action $A(u, x)$ that minimizes their own expected loss. This reduces to a pointwise minimization for each observed state (u, x) :

$$A(u, x) = \arg \min_{a \in \{\emptyset, x\}} \mathbb{E}[\ell(u, \hat{y}(M)) \mid A(u, x) = a]. \quad (1)$$

Anticipating the Sender's best response, the Receiver must now optimize the choice of $\hat{y}$ to minimize their own total expected loss, achieving the strategic risk R_S :

$$R_S := \min_{\hat{y} \in \mathcal{R}} \mathbb{E}[\ell(Y, \hat{y}(M))]. \quad (2)$$

Such a pair of strategies $(A, \hat{y})$ constitutes a **Stackelberg equilibrium**.

To evaluate the strategic gain, we compare this equilibrium to a mandatory disclosure regime (we also refer to this as the **vanilla** setting). Under mandatory disclosure, the Sender is forced to reveal features ($A \equiv x$), which leads to the Receiver’s optimal strategy $\hat{y}_V$. The corresponding vanilla Risk is:

$$R_V := \mathbb{E}[\ell(Y, \hat{y}_V(M))]. \quad (3)$$

Definition 1 (Strategic Advantage). *A **strategic advantage** exists if permitting voluntary disclosure yields a strictly lower expected loss for the Receiver than the mandatory regime, i.e., $R_S < R_V$.*

3 MAIN RESULTS

In this section, we characterize the conditions under which a strategic advantage arises across different learning settings. Throughout our analysis, we adopt the shorthand notation $\hat{y}(\emptyset)$ to denote the Receiver’s prediction when features are withheld (also called default prediction), $\hat{y}(x)$ when features are revealed, $\hat{y}(M)$ when evaluating on the observed message M , and $\hat{y}_V(M)$ for the vanilla predictor’s output on M .

Firstly, we analyze the general regression problem under the assumption of a known environment, deriving distinct necessary and sufficient conditions for the emergence of a strategic advantage. Secondly, we turn to the binary classification problem—also assuming a known environment—and establish the necessary and sufficient conditions for strategic advantage; notably, for classification, these conditions are found to be equivalent. Thirdly, we consider an online learning setting where the environment is unknown, and the Receiver must interact repeatedly with the Sender. In this case, we propose online optimization algorithms under partial feature observation and provide theoretical convergence guarantees for the receiver’s learning process. Detailed proofs for all results in this section are deferred to the Postponed Proofs section 6.

3.1 Strategic Advantage Condition in Regression

We now sharpen our problem formulation (Section 2) for the regression setup and only mention the difference.

The outcome space $\mathcal{Y} = \mathbb{R}$ is the space of real numbers and the loss function $\ell(y, \hat{y}(M)) = (y - \hat{y}(M))^2$ is mean squared error (MSE). For this regression formulation, we explicitly assume that all random variables (U, X, Y) have zero mean everywhere. In the mandatory disclosure setup, the Sender always chooses $A = x$. For a linear Receiver strategy $\hat{y}_V(\emptyset) = 0$ and $\hat{y}_V(x) = b^\top x$, we characterize the resulting risk in the following lemma.

Lemma 1. *In the mandatory disclosure setup, for a Receiver strategy parameterized by $b \in \mathbb{R}^d$, the expected risk is*

$$R_V(b, q) = \text{Var}(Y) - (1 - q) \text{Var}(b_Y^\top X) + (1 - q) \text{Var}((b - b_Y)^\top X), \quad (4)$$

where $b_Y = \text{Var}(X)^{-1} \text{Cov}(X, Y)$. The optimal vanilla risk is achieved at $b = b_Y$, yielding

$$R_V^*(q) = \text{Var}(Y) - (1 - q) \text{Var}(b_Y^\top X). \quad (5)$$

Remark 1. *Note that $b_Y = \text{Var}(X)^{-1} \text{Cov}(X, Y)$ is the optimal weight vector for ordinary least squares (OLS) regression of Y on X .*

In the voluntary disclosure setup, for a given Receiver strategy $\hat{y}(M)$, we consider the linear form $\hat{y}(\emptyset) = 0$ and $\hat{y}(x) = b^\top X$, where $b \in \mathbb{R}^d$. For a fixed b , we define the following scalar fields:

$$F(x, u) := (2u - b^\top x)b^\top x, \quad (6)$$

$$G(x, y) := (2y - b^\top x)b^\top x. \quad (7)$$

The Sender's best response action is then characterized by the following lemma.

Lemma 2. *In the voluntary disclosure setup, for a given receiver strategy $\hat{y}(M)$, the Sender's best response action is given by*

$$A(x, u) = x \iff F(x, u) > 0. \quad (8)$$

Based on this best response, we derive the Receiver's strategic risk R_S in the next lemma.

Lemma 3. *Given a Receiver strategy $\hat{y}(M)$ parameterized by b , the resulting strategic risk is*

$$R_S(b, q) = \text{Var}(Y) - (1 - q)\mathbb{E}[G(X, Y)\mathbb{I}\{F(X, U) > 0\}], \quad (9)$$

where $\mathbb{I}\{\cdot\}$ denotes the indicator function.

Remark 2. *The Stackelberg equilibrium risk is defined as the minimum strategic risk over the parameter space:*

$$R_S^*(q) := \min_b R_S(b, q).$$

Remark 3. *The function $F(x, u)$ serves as the decision rule for the Sender. The function $G(x, y)$ represents the Receiver's gain when the feature vector x is successfully received through the noisy channel (i.e., the signal is not lost to noise) relative to when the feature vector is lost.*

The strategic advantage condition is $R_V^*(q) - R_S^*(q) > 0$. We provide a sufficiency condition for this inequality in the following theorem.

Theorem 1 (Sufficiency Condition). *A strategic advantage exists if there exists $b \in \mathbb{R}^d$ such that:*

$$\begin{aligned} \|U - Y\|_2 &< \frac{\mathbb{E}[|G(X, Y)|] - \text{Var}(b_Y^\top X)}{4\sqrt{\text{Var}(b_Y^\top X)}} \\ &\quad - \frac{\text{Var}((b - b_Y)^\top X)}{4\sqrt{\text{Var}(b_Y^\top X)}}, \end{aligned} \quad (10)$$

where $\|U - Y\|_2 = \sqrt{\text{Var}(U) + \text{Var}(Y) - 2\text{Cov}(U, Y)}$, $b_Y = \text{Var}(X)^{-1} \text{Cov}(X, Y)$, and $G(X, Y) = (2Y - b^\top X)b^\top X$.

Beyond sufficiency, we can also identify a necessary condition for a strategic advantage to be possible. We define a strategic disadvantage when $R_S^*(q) = R_V^*(q)$; therefore, a necessary condition for an advantage must be the complement of this state.

Theorem 2 (Necessary Condition). *A necessary condition for a strategic advantage to exist is:*

$$\begin{aligned} \max_b \left(\mathbb{E}[|G(X, Y)|] - \text{Var}(b_Y^\top X) \right. \\ \left. - \text{Var}((b - b_Y)^\top X) \right) > 0. \end{aligned} \quad (11)$$

Remark 4. The necessary condition in (11) is equivalent to the requirement that the sufficiency condition in (10) be non-vacuous. Since the left-hand side of (10) is non-negative, the condition in (11) is equivalent to the numerator on the right-hand side being positive (as the denominator is always positive), thereby allowing for the existence of a non-empty range for $\|U - Y\|_2$.

Remark 5. Both the necessary and sufficient conditions are independent of the channel noise parameter q . This highlights that channel noise only controls the magnitude of the strategic advantage, not whether it can be attained. Additionally, both theorems require the bounds to hold for the best possible parameter b . Consequently, if the conditions are satisfied evaluated simply at the vanilla optimal weight b_Y , they necessarily hold under optimal b .

Remark 6. To demonstrate why the numerator of the right-hand side in Theorem 1 is strictly bounded when taking $\max_b$, we analyze the algebraic structure of the gain function $G(X, Y)$. The numerator is defined as $\mathbb{E}[|G(X, Y)|] - \text{Var}(b_Y^\top X) - \text{Var}((b - b_Y)^\top X)$. From the properties of the ordinary least squares predictor, we use the identity $\mathbb{E}[G(X, Y)] = \text{Var}(b_Y^\top X) - \text{Var}((b - b_Y)^\top X)$. We can use this to rewrite the expression by substituting the negative variance penalty:

$$\begin{aligned} \text{Numerator}(b) &= \mathbb{E}[|G(X, Y)|] + \mathbb{E}[G(X, Y)] - 2 \text{Var}(b_Y^\top X) \\ &= 2\mathbb{E}[\max(G(X, Y), 0)] - 2 \text{Var}(b_Y^\top X). \end{aligned}$$

By definition, the gain function is $G(X, Y) = (2Y - b^\top X)b^\top X$. Expanding and completing the square with respect to $b^\top X$ yields $G(X, Y) = Y^2 - (Y - b^\top X)^2$. This parabolic form guarantees that $G(X, Y) \leq Y^2$ for any arbitrary vector b . Consequently, it holds that $\max(G(X, Y), 0) \leq Y^2$. Because Y is a zero-mean random variable, $\mathbb{E}[Y^2] = \text{Var}(Y)$. Taking the expectation provides a definitive, finite upper bound:

$$\max_b \text{Numerator}(b) \leq 2 \text{Var}(Y) - 2 \text{Var}(b_Y^\top X).$$

Since $\text{Var}(Y)$ and the variance of the optimal vanilla predictor $\text{Var}(b_Y^\top X)$ are fixed, finite constants for any given environment, the supremum over $b \in \mathbb{R}^d$ is securely bounded and cannot explode to infinity.

To numerically verify these theoretical results, we simulate a regression environment using a family of zero-mean Gaussian distributions for (X, Y, U) with unit marginal variances. We fix $\text{Cov}(X, Y) = \text{Cov}(X, U) = 0.2$ across this family, while $\text{Cov}(Y, U)$ serves as the varying parameter. To explore the parameter space, we sweep the channel noise probability $q \in [0, 1]$ and the alignment $\text{Cov}(Y, U) \in [-1, 1]$. At each point on the grid, the Receiver's risk is minimized over a range of weights b to calculate the strategic risk R_S . As demonstrated in Fig. 1, the analytical boundary reliably delineates the regions of strategic advantage.

3.1.1 Gaussian Specialization

We now specialize the general regression results to the case where (U, X, Y) are jointly Gaussian with zero mean. Under this distributional assumption, the sufficiency condition of Theorem 1 admits a closed-form expression in terms of the covariance parameters.

Theorem 3 (Gaussian Sufficiency Condition). *Let $(U, X, Y) \sim \mathcal{N}(0, \Sigma)$ be jointly Gaussian with covariance matrix Σ . By making a specific substitution for the Receiver predictor $b = 2\Sigma_{XX}^{-1}\Sigma_{XU}$, a strategic advantage exists if*

$$2(b_Y^\top \Sigma_{XU} - \sigma_Z^2) + \frac{8\kappa\sigma_Z}{s\sqrt{2\pi}} e^{-\alpha^2/2} > \Sigma_{XY}^\top \Sigma_{XX}^{-1} \Sigma_{XY}. \quad (12)$$

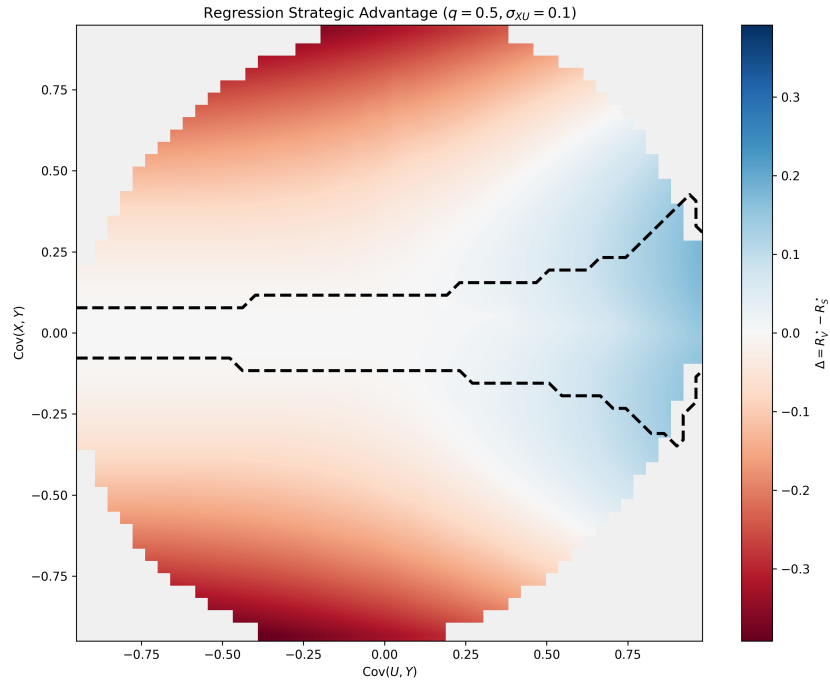


Figure 1: Heatmap of the strategic advantage ($R_V^* - R_S^*$) for regression ($q = 0.5$). Horizontal axis: $\text{Cov}(U, Y)$, Vertical axis: $\text{Cov}(X, Y)$. The black dotted line represents the theoretical sufficiency condition (Theorem 1). Parameters were chosen based on sufficiency discovery ($\text{Cov}(X, U) = 0.3$). Blue regions indicate parameter regimes where Voluntary disclosure is superior.

3.2 Strategic Advantage Condition in Classification

We now sharpen our problem formulation (Section 2) for the classification setup and only mention the difference.

The outcome space $\mathcal{Y} = \{-1, +1\}$ is the space of discrete labels and the loss function $\ell(y, \hat{y}) = \mathbb{I}\{\hat{y} \neq y\}$ is the 0-1 loss. In the mandatory disclosure setup, the Sender always chooses $A = X$. For a Receiver strategy defined by $\hat{y}_V(\emptyset) = \text{sign}(\mathbb{E}[Y])$ and $\hat{y}_V(x) = \text{sign}(\mathbb{E}[Y|X = x])$, we characterize the vanilla resulting risk in the following lemma.

Lemma 4 (Classification Vanilla Risk). *In the mandatory disclosure setup, the expected vanilla risk with noisy channel q is*

$$\begin{aligned} R_V^*(q) &= q\mathbb{P}(Y = -\text{sign}(\mathbb{E}[Y])) \\ &\quad + (1 - q)\mathbb{E}\left[\frac{1 - |\eta(X)|}{2}\right], \end{aligned} \quad (13)$$

where $\eta(X) = \mathbb{E}[Y|X = x]$.

In the voluntary disclosure setup, the Receiver commits to a default prediction $\hat{y}(\emptyset) = i \in \{-1, +1\}$, which we refer to as the default option. A Sender with preference u observes (x, u) and chooses whether to disclose x . Under the 0-1 loss, the Sender withholds the feature if the default option matches their preference ($u = i$) and discloses it if $u = -i$ to potentially shift the prediction. If the Receiver observes the message $M = X$, the disclosure reveals that $u = -i$, prompting the Receiver to use the optimal posterior predictor $\hat{y}(x) = \text{sign}(\mathbb{E}[Y | X = x, U = -i])$.

Because the optimum Receiver strategies are governed entirely by their choice of the default option i , there are only two possible strategies in the voluntary setup. Let us analyze each of them. Assuming without loss of generality that $\mathbb{E}[Y] > 0$, we denote the strategy where the default option matches the sign of the prior mean as $\mathcal{S}_1$ (i.e., $i = 1$) and the alternative strategy as $\mathcal{S}_{-1}$. We characterize the resulting strategic risk for both of these cases in the following lemma.

Lemma 5 (Strategic Classification Risk). *For a default prediction $i \in \{-1, 1\}$, let $\pi_i = \mathbb{P}(U = i)$ and $\eta_{-i}(X) = \mathbb{E}[Y | X, U = -i]$. The strategic risk $R_{\mathcal{S}_i}^*(q)$ is:*

$$\begin{aligned} R_{\mathcal{S}_i}^*(q) &= \pi_i\mathbb{P}(Y = -i | U = i) \\ &\quad + \pi_{-i}\left[(1 - q)\mathbb{E}\left[\frac{1 - |\eta_{-i}(X)|}{2}\right] \right. \\ &\quad \left. + q\mathbb{P}(Y = -i | U = -i)\right]. \end{aligned} \quad (14)$$

To determine the Stackelberg equilibrium, the Receiver must select the strategy $\mathcal{S}_i$ that has the minimum risk, i.e., $R_{\mathcal{S}_i}^*(q) = \min_i R_{\mathcal{S}_i}^*(q)$. Since the channel noise parameter q acts linearly on the risk functions, we can characterize the optimal policy based on the value of q . By substituting the risk expressions from Lemmas 4 and 5, we formally characterize both the Stackelberg equilibrium and the conditions for strategic advantage.

Theorem 4 (Stackelberg Equilibrium and Strategic Advantage). *Assume $\mathbb{E}[Y] > 0$. At the maximum noise limit ($q = 1$), $\mathcal{S}_1$ is strictly better (i.e., $R_{\mathcal{S}_1}^*(1) < R_{\mathcal{S}_{-1}}^*(1)$). At $q = 0$, the optimal strategy depends on the underlying distributions, leading to two possible scenarios:*

1. If $R_{\mathcal{S}_1}^*(0) \leq R_{\mathcal{S}_{-1}}^*(0)$, then $\mathcal{S}_1$ is the Stackelberg strategy for all $q \in [0, 1]$. The strategic advantage condition $R_V^*(q) > R_S^*(q)$ simplifies to:

$$(1 - q) \left[\mathbb{E} \left[\frac{1 - |\eta(X)|}{2} \right] - \pi_{-1} \mathbb{E} \left[\frac{1 - |\eta_{-1}(X)|}{2} \right] - \pi_1 \mathbb{P}(Y = -1 \mid U = 1) \right] > 0. \quad (15)$$

2. If $\mathcal{S}_{-1}$ is better at $q = 0$ (i.e., $R_{\mathcal{S}_1}^*(0) > R_{\mathcal{S}_{-1}}^*(0)$), there exists an intermediate threshold $q^* \in [0, 1]$ given by

$$q^* = \frac{R_{\mathcal{S}_1}^*(0) - R_{\mathcal{S}_{-1}}^*(0)}{(R_{\mathcal{S}_{-1}}^*(1) - R_{\mathcal{S}_{-1}}^*(0)) - (R_{\mathcal{S}_1}^*(1) - R_{\mathcal{S}_1}^*(0))}, \quad (16)$$

such that for $q \geq q^*$, $\mathcal{S}_1$ is the Stackelberg strategy and the advantage condition is identical to (15). For $q < q^*$, $\mathcal{S}_{-1}$ is the Stackelberg strategy, and the strategic advantage requires:

$$(1 - q) \left[\mathbb{E} \left[\frac{1 - |\eta(X)|}{2} \right] - \pi_1 \mathbb{E} \left[\frac{1 - |\eta_1(X)|}{2} \right] \right] + q \left[\mathbb{P}(Y = -1) - \pi_1 \mathbb{P}(Y = 1 \mid U = 1) \right] > \pi_{-1} \mathbb{P}(Y = 1 \mid U = -1). \quad (17)$$

Remark 7. In case (1) of Theorem 4, the condition in (15) is independent of q : the channel noise parameter only scales the magnitude of the advantage via the $(1 - q)$ factor, analogous to the regression setting. In contrast, in case (2), the condition in (17) explicitly depends on q , meaning the channel noise governs both the magnitude and the direction of the strategic advantage.

To empirically verify these theoretical results, we simulate the classification regime over a family of distributions. We assume discrete labels $(Y, U) \in \{-1, 1\}^2$ sampled from a joint distribution with fixed marginals $P(Y = 1) = 0.6$ and $P(U = 1) = 0.5$, while we vary $\text{Cov}(Y, U)$ as the varying parameter. The features follow a Gaussian mixture model $X \mid Y, U \sim Y\mu_Y + U\mu_U + \mathcal{N}(0, 1)$, with $\mu_Y = 1.5$ and $\mu_U = 0.5$. To explore the parameter space, we sweep the channel noise probability $q \in [0, 1]$ and the alignment $\text{Cov}(Y, U) \in [-1, 1]$. At each point on the grid, the strategic risk $R_S^*(q)$ is evaluated by selecting the optimal Receiver strategy. As demonstrated in Fig. 2, the analytical boundary reliably delineates the regions of strategic advantage.

4 LEARNING PROBLEM

In the previous sections, we analyzed the optimal disclosure regimes under the assumption that the underlying environment parameters—specifically, the joint distribution of (X, U, Y) —are known to the Receiver. In this section, we transition from the decision problem to the learning problem, where the Receiver operates in an unknown environment. We consider two ways to pose the learning problem: static learning from a population sample and online learning through sequential interactions.

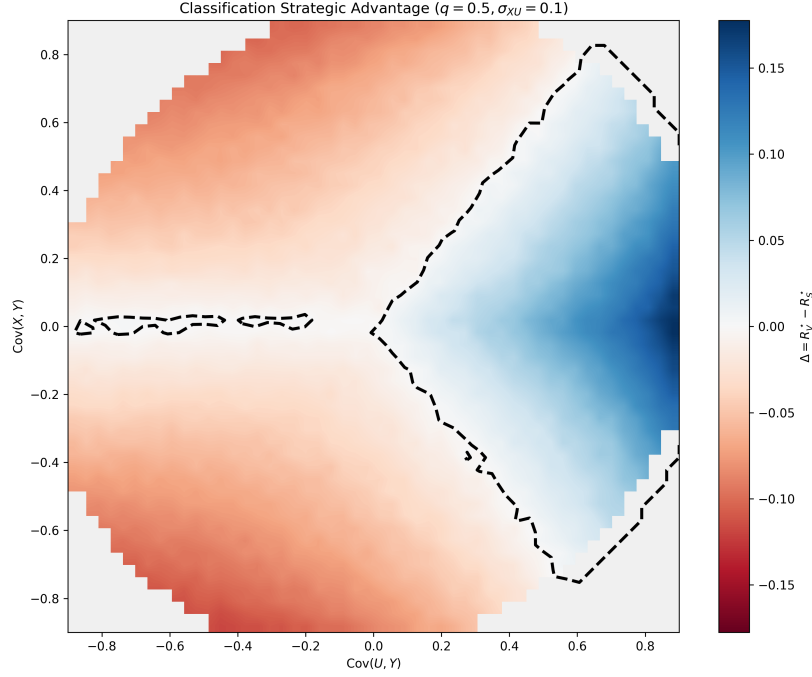


Figure 2: Heatmap of the strategic advantage for classification ($q = 0.5$). Horizontal axis: $\text{Cov}(U, Y)$, Vertical axis: $\text{Cov}(X, Y)$. The black dotted line represents the theoretical boundary ($\Delta = 0$). Blue regions indicate parameter regimes where Voluntary disclosure is superior.

Algorithm 1 Joint learning algorithm for strategy and disclosure regime

Require: Dataset $\mathcal{S} = \{(u_i, x_i, y_i)\}_{i=1}^n$ and strategy space

$\mathcal{R} := \{\hat{y}_b : \hat{y}_0 = 0, \hat{y}_x = b^\top x, \|b\|_\infty < C\}$ for some fixed constant $C > 0$.

▷ Solve empirical objectives

$\hat{b}_S \leftarrow \arg \min_b \hat{R}_S(b, q)$

$\hat{b}_V \leftarrow \arg \min_b \hat{R}_V(b, q)$

▷ Compute empirical strategic risk

$\hat{R}_S(\hat{b}_S, q) \leftarrow \frac{1}{n} \sum_{i=1}^n y_i^2 - (1 - q) \frac{1}{n} \sum_{i=1}^n ((2y_i - \hat{b}_S^\top x_i) \hat{b}_S^\top x_i \mathbb{I}\{(2u_i - \hat{b}_S^\top x_i) \hat{b}_S^\top x_i > 0\})$

▷ Compute empirical vanilla risk

$\hat{R}_V(\hat{b}_V, q) \leftarrow \frac{1}{n} \sum_{i=1}^n y_i^2 - (1 - q) \frac{1}{n} \sum_{i=1}^n (y_i - \hat{b}_V^\top x_i)^2$

if $\hat{R}_S(\hat{b}_S, q) < \hat{R}_V(\hat{b}_V, q)$ **then**

output $(\hat{b}_S, \text{voluntary})$

else

output $(\hat{b}_V, \text{mandatory})$

end if

4.1 Static Learning

In the static learning problem, the distribution of the random variable (U, X, Y) is unknown, but we have access to a finite number of samples from it. In this section, we present the following learning algorithm and establish bounds on its excess risk.

We now analyze the excess risk of the algorithm. We begin by introducing the necessary definitions and notation. The *empirical strategic advantage*, denoted as $\widehat{\Delta}(\mathcal{S})$, is defined as:

$$\widehat{\Delta}(\mathcal{S}) := \min_b \hat{R}_V(b, q) - \min_b \hat{R}_S(b, q).$$

Let $\mathcal{F}$ be a function class, and let $\mathcal{Z} := \{z_1, \dots, z_n\}$ be a collection of points in the domain of $\mathcal{F}$. The *empirical Rademacher complexity* of $\mathcal{F}$ over $\mathcal{Z}$, denoted by $\mathcal{R}(\mathcal{F}; \mathcal{Z})$, is defined as

$$\mathcal{R}(\mathcal{F}; \mathcal{Z}) := \mathbb{E}_\sigma \left[\sup_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^n \sigma_i f(z_i) \right],$$

where σ_i are i.i.d. Rademacher random variables. We denote by $d_{\text{VC}}(\mathcal{F})$ the VC dimension of the function class $\mathcal{F}$. For two function classes $\mathcal{F}$ and $\mathcal{G}$, the product class is defined as $\mathcal{F} \cdot \mathcal{G} := \{f \cdot g : f \in \mathcal{F}, g \in \mathcal{G}\}$, where $(f \cdot g)(z) = f(z)g(z)$ for all $z \in \mathbb{R}^d$. Furthermore, for a function $\phi : \mathbb{R} \rightarrow \mathbb{R}$, we define the composition class $\phi \circ \mathcal{G} := \{\phi \circ g : g \in \mathcal{G}\}$. If $\mathcal{F} = \phi \circ \mathcal{G}$, then for every $f \in \mathcal{F}$, there exists a $g \in \mathcal{G}$ such that $f(z) = \phi(g(z))$ for all $z \in \mathbb{R}^d$.

We establish generalization bounds for the vanilla risk and the strategic risk.

Lemma 6 (Vanilla Risk Generalization Bound). *Let $\mathcal{S} = \{(u_i, x_i, y_i)\}_{i=1}^n$ be a dataset of n i.i.d. samples. Let $\hat{y}_b \in \mathcal{R}$ be any receiver strategy. Let $\delta \in (0, 1)$. Then with probability at least $1 - \delta$, we have*

$$R_V(b, q) - \hat{R}_V(b, q) \leq \mathcal{O}\left(\frac{d^2 C^4}{\sqrt{n}} + C^2 \sqrt{\frac{\log(1/\delta)}{n}}\right).$$

Lemma 7 (Strategic Risk Generalization Bound). *Let $\mathcal{S} = \{(u_i, x_i, y_i)\}_{i=1}^n$ be a dataset of n i.i.d. samples. Let $\hat{y}_b \in \mathcal{R}$ be a receiver strategy. Let $\delta \in (0, 1)$. Then with probability at least $1 - \delta$, we have*

$$R_S(b, q) - \hat{R}_S(b, q) \leq \mathcal{O}\left(d^3 C^4 \sqrt{\frac{\log n}{n}} + C^2 \sqrt{\frac{\log(1/\delta)}{n}}\right). \quad (18)$$

From Lemma 6 and Lemma 7, we obtain the uniform generalization bound on the strategic advantage. Let $\Delta^* = R_V^*(q) - R_S^*(q)$ be the true strategic advantage.

Theorem 5. *Let $\mathcal{S} = \{(u_i, x_i, y_i)\}_{i=1}^n$ be a dataset of n i.i.d. samples drawn from the random variable (U, X, Y) over $\mathcal{Y} \times \mathcal{X} \times \mathcal{Y}$. Let $\delta \in (0, 1)$. Then with probability at least $1 - \delta$, we have*

$$|\Delta^* - \widehat{\Delta}(\mathcal{S})| \leq \mathcal{O}\left(d^3 C^4 \sqrt{\frac{\log n}{n}} + C^2 \sqrt{\frac{\log(1/\delta)}{n}}\right).$$

We now establish the excess risk bound on the joint selection learning algorithm.

Theorem 6. *Consider Algorithm 1 and dataset $\mathcal{S}$. Let $\delta \in (0, 1)$. Then with probability at least $1 - \delta$, the expected excess risk of the algorithm relative to the optimal disclosure regime is bounded by*

$$\text{Excess Risk} \leq \mathcal{O}\left(d^3 C^4 \sqrt{\frac{\log n}{n}} + C^2 \sqrt{\frac{\log(1/\delta)}{n}}\right).$$

The proofs of all the results in this subsection are provided in Appendix 6.3.

4.2 Online Learning Framework

In the online learning setting, we study a sequential interaction over T rounds between the Receiver (the learner) and a sequence of Strategic Senders. At each round t , a sender is drawn from a population and decides whether to reveal their private features to the receiver based on the receiver's published model.

Assumptions: We assume a noiseless channel ($q = 0$). All environment variables are bounded by a constant $C > 0$:

- Feature vectors $x_t \in \mathcal{X} \subset \mathbb{R}^d$ satisfy $\|x_t\|_\infty \leq C$.
- Ground truth labels $y_t \in \mathcal{Y} \subset \mathbb{R}$ satisfy $|y_t| \leq C$.
- Private preferences $u_t \in \mathcal{U} \subset \mathbb{R}$ satisfy $|u_t| \leq C$.
- Model weights $b_t \in \mathcal{B} \subset \mathbb{R}^d$ satisfy $\|b_t\|_\infty \leq C$.

Furthermore, we assume a bounded alignment error between the sender's private preference U and the ground truth Y , measured by the L^2 -norm: $\|U - Y\|_{L^2} := \sqrt{\mathbb{E}[(U - Y)^2]} \leq \epsilon$ for some $\epsilon > 0$.

4.2.1 The Sequential Round Structure

For each round $t = 1, 2, \dots, T$, the interaction proceeds as follows:

1. **Model Commitment:** The receiver publishes a parameter vector $b_t \in \mathcal{B}$ and commits to a prediction rule $\hat{y}_{b_t}(x) = b_t^\top x$ for revealed features, and $\hat{y}_{b_t}(\emptyset) = 0$ for withheld features.
2. **Nature's Sample:** Nature draws a triplet $z_t := (u_t, x_t, y_t) \sim \mathcal{P}$ from an underlying distribution $\mathcal{P}$.
3. **Sender's Strategic Decision:** The sender evaluates the relative advantage of revealing features x_t over withholding them under the current model b_t . The relative loss of revelation is:

$$F_{b_t}(z_t) = \text{Loss}(\text{Withhold}) - \text{Loss}(\text{Reveal}) = (2u_t - b_t^\top x_t)b_t^\top x_t \quad (19)$$

The sender chooses to reveal features with probability $A_t \in \{0, 1\}$ sampled as:

$$A_t \sim \text{Bernoulli}(\psi(F_{b_t}(z_t))) \quad (20)$$

where $\psi(s) = 1/(1 + e^{-\kappa s})$ is a smooth, monotonically increasing activation function (e.g., a logistic/sigmoid function with sensitivity κ). Note that ψ is L_ψ -Lipschitz with $L_\psi = \kappa/4$.

4. **Observation and Loss:** The receiver observes the outcome (y_t, A_t) and the feature vector x_t (if $A_t = 1$). The receiver then incurs the instantaneous quadratic loss:

$$\ell_t(b_t; z_t, A_t) = (1 - A_t)y_t^2 + A_t(y_t - b_t^\top x_t)^2 \quad (21)$$

4.2.2 Strategic Population Risk

The goal of the receiver is to minimize the *Strategic Population Risk*, defined as the expected loss over the population distribution and revelation outcomes:

$$\mathcal{L}(b) = \mathbb{E}_{z_t \sim \mathcal{P}, A_t \sim \psi} [\ell_t(b; z_t, A_t)] \quad (22)$$

By taking the expectation over the revelation indicator A_t , we can rewrite the population risk as:

$$\mathcal{L}(b) = \mathbb{E}_{z_t} [y_t^2 - \psi(F_b(z_t)) \cdot G_b(z_t)] \quad (23)$$

where $G_b(z_t) = (2y_t - b^\top x_t)b^\top x_t$ represents the reduction in squared error relative to the label y_t .

4.2.3 Non-Convexity and Distributional Strong Convexity

It is critical to note that the Strategic Population Risk $\mathcal{L}(b)$ is broadly **non-convex**. This non-convexity arises because the revelation probability $\psi(F_b(z_t))$ is a non-linear function intricately coupled with the model parameter b .

We impose a **Distributional Strong Convexity** assumption: we assume there exists a local minimum b^* (closest to the initialization b_1) around which the distribution $\mathcal{P}$ guarantees the Hessian of the population risk is strictly positive definite:

$$\nabla^2 \mathcal{L}(b^*) \succeq \lambda I \quad \text{for some } \lambda > 0. \quad (24)$$

This structural condition ensures that, at least locally, there is a prominent basin of attraction pointing toward the minimum wherein the risk behaves convexly.

4.3 Learning with Full Feedback

In the ideal benchmark where features (x_t, u_t, y_t) are observed regardless of the disclosure action A_t , the receiver employs Online Gradient Descent (OGD) to minimize the Strategic Population Risk. The gradient $\nabla \mathcal{L}(b)$ is derived as follows:

$$\nabla \mathcal{L}(b) = -2\mathbb{E}_{z_t} \left[\left(\psi'(F_b(z_t))(u_t - b^\top x_t)G_b(z_t) + \psi(F_b(z_t))(y_t - b^\top x_t) \right) x_t \right] \quad (25)$$

The receiver performs the unbiased stochastic gradient update: $b_{t+1} = \Pi_{\mathcal{B}}(b_t - \eta_t g_t)$, where g_t is the instantaneous gradient.

Theorem 7 (Full Feedback Convergence). *Assume the environment fulfills the Distributional Local Convexity and regularity conditions (bounded gradients $\|g_t\| \leq G$ and bounded domain $\|b_1 - b^*\| \leq D$). Running OGD with learning rate $\eta = \frac{D}{G\sqrt{T}}$ yields the expected average excess strategic risk bound:*

$$\frac{1}{T} \sum_{t=1}^T \mathbb{E}[\mathcal{L}(b_t) - \mathcal{L}(b^*)] \leq \mathcal{O}\left(\frac{1}{\sqrt{T}}\right). \quad (26)$$

The proof of Theorem 7 is provided in Appendix 6.4.

4.4 Learning with Partial Feedback

In realistic settings, the bliss point u_t is strictly private, and features x_t are missing when the sender chooses to withhold ($A_t = 0$). The receiver approximates $\mathcal{L}(b)$ with the *Surrogate Risk* $\tilde{\mathcal{L}}(b)$, substituting u_t with the observed label y_t :

$$\tilde{\mathcal{L}}(b) = \mathbb{E}_{z_t} [y_t^2 - \psi(G_b(z_t)) \cdot G_b(z_t)] \quad (27)$$

When updating b_t , the learner uses a surrogate stochastic gradient $\tilde{g}_t$. Since u_t is missing, the learner assumes $u_t \approx y_t$. If $A_t = 0$, x_t is unobserved and is substituted with its mean value (here 0). The surrogate update is $b_{t+1} = \Pi_{\mathcal{B}}(b_t - \eta_t \tilde{g}_t)$ where:

$$\tilde{g}_t = \begin{cases} -2 [\psi'(G_{b_t}(z_t))(y_t - b_t^\top x_t)G_{b_t}(z_t) + \psi(G_{b_t}(z_t))(y_t - b_t^\top x_t)] x_t & \text{if } A_t = 1 \\ 0 & \text{if } A_t = 0 \end{cases} \quad (28)$$

Theorem 8 (Partial Feedback Convergence). *Let $\tilde{b}^*$ denote the local minimum of $\tilde{\mathcal{L}}(b)$. Assuming local λ -strong convexity and L -smoothness of the surrogate risk, and an expected revelation probability bounded away from zero ($p_{\min} > 0$), the total expected average excess strategic risk is bounded by:*

$$\frac{1}{T} \sum_{t=1}^T \mathbb{E}[\mathcal{L}(b_t) - \mathcal{L}(b^*)] \leq \mathcal{O}\left(\frac{1}{\sqrt{T}}\right) + \max\{\mathcal{O}(\epsilon^2), \mathcal{O}(\epsilon)\}. \quad (29)$$

where the first term decays with time and the second term represents the structural bias due to unobserved preferences ($\|U - Y\|_{L^2} \leq \epsilon$).

The proof of Theorem 8 is provided in Appendix 6.4.

5 CONCLUSION

In this paper, we have provided sufficiency and necessity conditions for strategic advantages in both regression and classification settings. These conditions highlight the role of misalignment and the noisy channel parameter q in the receiver's choice between forced and optional revelation modes. Furthermore, we explored the learning problem in both static and online settings, proposing online optimization algorithms for strategic regression under partial feature observation and establishing sublinear convergence bounds for the learning process. Future work includes the extension to multi-sender environments.

6 POSTPONED PROOFS

Here we present the proof of all results we skipped in the Section 3.

6.1 Proofs for Strategic Advantage Condition for Regression

This section provides the complete derivations for the general regression risk formulations and the subsequent sufficiency and necessity conditions.

Proof of Lemma 1. We compute the risk in the mandatory disclosure setting for a linear predictor $b \in \mathbb{R}^d$ with $\hat{y}_V(\emptyset) = 0$ and $\hat{y}_V(x) = b^\top x$. The risk is given by:

$$R_V(b, q) = q\mathbb{E}[Y^2] + (1 - q)\mathbb{E}[(Y - b^\top X)^2].$$

Since all variables have mean 0, $\text{Var}(Y) = \mathbb{E}[Y^2]$. We now examine the term $\mathbb{E}[(Y - b^\top X)^2]$. We add and subtract $b_Y^\top X$ inside the expectation, where $b_Y = \text{Var}(X)^{-1} \text{Cov}(X, Y)$:

$$\begin{aligned}\mathbb{E}[(Y - b^\top X)^2] &= \mathbb{E}[(Y - b_Y^\top X)^2] \\ &\quad + \mathbb{E}[((b_Y - b)^\top X)^2] \\ &\quad + 2\mathbb{E}[(Y - b_Y^\top X)(b_Y - b)^\top X].\end{aligned}$$

By the orthogonality principle of OLS, $\mathbb{E}[(Y - b_Y^\top X)X] = 0$, so the cross-term vanishes. Thus,

$$\mathbb{E}[(Y - b^\top X)^2] = \text{Var}(Y) - \text{Var}(b_Y^\top X) + \text{Var}((b - b_Y)^\top X).$$

Substituting back into the risk yields (4). The optimal strategy minimizes $\text{Var}((b - b_Y)^\top X)$ by choosing $b = b_Y$. $\square$

Proof of Lemma 2. Given a realization (x, u) , the Sender selects the action $A \in \{\emptyset, x\}$ minimizing $(u - \hat{y}(M))^2$. The expected loss from revealing is $q u^2 + (1 - q)(u - b^\top x)^2$, while from withholding it is u^2 . The Sender prefers revealing if:

$$u^2 > q u^2 + (1 - q)(u - b^\top x)^2.$$

Simplifying yields $(b^\top x)(2u - b^\top x) > 0$, which is $F(x, u) > 0$. $\square$

Proof of Lemma 3. For a fixed parameter b , the strategy of the Receiver is $\hat{y}_V(\emptyset) = 0$ for an empty message $M = \emptyset$ and $\hat{y}_V(x) = b^\top x$ for a successfully received feature vector $M = X$.

According to Lemma 2, the Sender reveals the feature vector x if and only if $F(X, U) > 0$. Using the law of total expectation, we decompose the strategic risk $R_S(b, q) = \mathbb{E}[\ell(Y, \hat{y}(M))]$ into the two cases of F :

$$\begin{aligned}R_S(b, q) &= \mathbb{E}[\mathbb{I}\{F \leq 0\} \ell(Y, \hat{y}(\emptyset)) \\ &\quad + \mathbb{I}\{F > 0\} (q \ell(Y, \hat{y}(\emptyset)) + (1 - q) \ell(Y, \hat{y}(X)))] \\ &= \mathbb{E}[\mathbb{I}\{F \leq 0\} Y^2 \\ &\quad + \mathbb{I}\{F > 0\} (q Y^2 + (1 - q)(Y - b^\top X)^2)].\end{aligned}$$

Applying the identity $\mathbb{I}\{F \leq 0\} = 1 - \mathbb{I}\{F > 0\}$:

$$\begin{aligned}R_S(b, q) &= \mathbb{E}[Y^2] - (1 - q) \mathbb{E}[\mathbb{I}\{F > 0\} (Y^2 - (Y - b^\top X)^2)] \\ &= \text{Var}(Y) - (1 - q) \mathbb{E}[\mathbb{I}\{F > 0\} (2Y(b^\top X) \\ &\quad - (b^\top X)^2)],\end{aligned}$$

where $\text{Var}(Y) = \mathbb{E}[Y^2]$ follows from the zero-mean assumption. Substituting $G(X, Y) = (2Y - b^\top X)b^\top X$ yields:

$$R_S(b, q) = \text{Var}(Y) - (1 - q) \mathbb{E}[\mathbb{I}\{F > 0\} G(X, Y)],$$

which concludes the proof. $\square$

Proof of Theorem 1 (Sufficiency Condition). We seek condition such that $R_S^*(q) - R_V^*(q) < 0$. From Lemmas 1 and 3:

$$\begin{aligned}R_S^*(q) - R_V^*(q) &= \text{Var}(Y) - (1 - q) \mathbb{E}[G \mathbb{I}\{F > 0\}] \\ &\quad - (\text{Var}(Y) - (1 - q) \text{Var}(b_Y^\top X)).\end{aligned}$$

Using $\mathbb{I}\{F > 0\} = 1 - \mathbb{I}\{F \leq 0\}$, we have $\mathbb{E}[G\mathbb{I}\{F > 0\}] = \mathbb{E}[G] - \mathbb{E}[G\mathbb{I}\{F \leq 0\}]$.

To compute $\mathbb{E}[G]$, recall $G(X, Y) = (2Y - b^\top X)b^\top X$. Since $\mathbb{E}[X] = 0$, $\text{Var}(b^\top X) = \mathbb{E}[(b^\top X)^2]$. Furthermore, the optimal OLS weights $b_Y = \mathbb{E}[XX^\top]^{-1}\mathbb{E}[XY]$ satisfy $\mathbb{E}[Y(b^\top X)] = b^\top \mathbb{E}[XY] = b^\top \text{Var}(X)b_Y = \text{Cov}(b^\top X, b_Y^\top X)$. Thus,

$$\begin{aligned}\mathbb{E}[G] &= 2\mathbb{E}[Y(b^\top X)] - \mathbb{E}[(b^\top X)^2] \\ &= 2b^\top \text{Var}(X)b_Y - \text{Var}(b^\top X).\end{aligned}$$

From the identity $\text{Var}((b - b_Y)^\top X) = \text{Var}(b^\top X) + \text{Var}(b_Y^\top X) - 2b^\top \text{Var}(X)b_Y$, we obtain:

$$\mathbb{E}[G] = \text{Var}(b_Y^\top X) - \text{Var}((b - b_Y)^\top X).$$

Substituting this into the risk difference we get:

$$R_S^*(q) - R_V^*(q) = (1-q)(\text{Var}((b - b_Y)^\top X) + \mathbb{E}[G\mathbb{I}\{F \leq 0\}]).$$

We upper bound $\mathbb{E}[G\mathbb{I}\{F \leq 0\}]$ by decomposing:

$$\begin{aligned}\mathbb{E}[G\mathbb{I}\{F \leq 0\}] &= \mathbb{E}[G\mathbb{I}\{G \leq 0\}] \\ &\quad + \mathbb{E}[G(\mathbb{I}\{F \leq 0\} - \mathbb{I}\{G \leq 0\})] \\ &= -\frac{1}{2}(\mathbb{E}[|G|] - \mathbb{E}[G]) \\ &\quad + \mathbb{E}[G\mathbb{I}\{\text{sign}(F) \neq \text{sign}(G)\}].\end{aligned}$$

When $\text{sign}(F) \neq \text{sign}(G)$, we have $|G| \leq |G - F| = 2|Y - U||b^\top X|$. By Cauchy-Schwarz:

$$\begin{aligned}\mathbb{E}[G\mathbb{I}\{\text{sign}(F) \neq \text{sign}(G)\}] &\leq \mathbb{E}[2|Y - U||b^\top X|] \\ &\leq 2\|U - Y\|_2 \sqrt{\text{Var}(b_Y^\top X)}.\end{aligned}$$

Setting upper bound to be negative, (which would imply $R_S^*(q) - R_V^*(q) < 0$), we get:

$$\begin{aligned}&\text{Var}((b - b_Y)^\top X) \\ &\quad - \frac{1}{2}(\mathbb{E}[|G|] - \text{Var}(b_Y^\top X) \\ &\quad \quad + \text{Var}((b - b_Y)^\top X)) \\ &\quad + 2\|U - Y\|_2 \sqrt{\text{Var}(b_Y^\top X)} < 0.\end{aligned}$$

Rearranging isolates $\|U - Y\|_2$, yielding (10). □

Proof of Theorem 2 (Necessary Condition). We prove the contrapositive: if (11) fails, then no strategic advantage can exist, i.e., $R_S(b, q) \geq R_V^*(q)$ for all b . From Lemma 1 and Lemma 3, we have:

$$R_S(b, q) - R_V^*(q) = (1-q)(\text{Var}((b - b_Y)^\top X) + \mathbb{E}[G\mathbb{I}\{F \leq 0\}]).$$

A strategic advantage requires $R_S(b, q) - R_V^*(q) < 0$ for some b . We can lower bound the term $\mathbb{E}[G\mathbb{I}\{F \leq 0\}]$ purely algebraically.

If $G \leq 0$, we have $G\mathbb{I}\{F \leq 0\} \geq G = G\mathbb{I}\{G \leq 0\}$. If $G > 0$, we have $G\mathbb{I}\{F \leq 0\} \geq 0 = G\mathbb{I}\{G \leq 0\}$. Therefore, pointwise $G\mathbb{I}\{F \leq 0\} \geq G\mathbb{I}\{G \leq 0\}$, which implies:

$$\mathbb{E}[G\mathbb{I}\{F \leq 0\}] \geq \mathbb{E}[G\mathbb{I}\{G \leq 0\}].$$

Substituting this inequality yields:

$$R_S(b, q) - R_V^*(q) \geq (1-q) \left(\text{Var}((b-b_Y)^\top X) + \mathbb{E}[G \mathbb{I}\{G \leq 0\}] \right).$$

Now $\mathbb{E}[G \mathbb{I}\{G \leq 0\}]$ depends only on (X, Y) and b , with U entirely eliminated. Using the identity $\mathbb{E}[G \mathbb{I}\{G \leq 0\}] = -\frac{1}{2}(\mathbb{E}[|G|] - \mathbb{E}[G])$ and $\mathbb{E}[G] = \text{Var}(b_Y^\top X) - \text{Var}((b-b_Y)^\top X)$:

$$\begin{aligned} R_S(b, q) - R_V^*(q) &\geq (1-q) \left(\text{Var}((b-b_Y)^\top X) \right. \\ &\quad \left. - \frac{1}{2}(\mathbb{E}[|G|] - \text{Var}(b_Y^\top X) \right. \\ &\quad \left. + \text{Var}((b-b_Y)^\top X)) \right). \end{aligned}$$

Multiplying by 2 and rearranging, $R_S(b, q) - R_V^*(q) < 0$ requires:

$$\mathbb{E}[|G|] - \text{Var}(b_Y^\top X) - \text{Var}((b-b_Y)^\top X) > 0.$$

If this fails for all b , then $R_S(b, q) - R_V^*(q) \geq 0$ for all b , meaning no strategic advantage is possible. Hence the necessary condition is that there exists some b satisfying (11) and thus

$$\max_b \mathbb{E}[|G|] - \text{Var}(b_Y^\top X) - \text{Var}((b-b_Y)^\top X) > 0.$$

□

6.1.1 Proof of Gaussian Specialization

Proof of Theorem 3. Under joint Gaussianity with zero mean, we define the following parameters for the analysis: $b_Y = \Sigma_{XX}^{-1} \Sigma_{XY}$ and $b_U = \Sigma_{XX}^{-1} \Sigma_{XU}$ as the respective OLS predictors of Y and U from X ; $\sigma_Z^2 = b_U^\top \Sigma_{XX} b_U = \Sigma_{XU}^\top \Sigma_{XX}^{-1} \Sigma_{XU}$, $\kappa = \text{Cov}(U, Y | X)$, $s^2 = \text{Var}(U | X)$, and $\alpha = -\mathbb{E}[U | X]/s = 0$ (since $\mathbb{E}[U] = 0$) as the normalized conditional mean of U .

We evaluate the strategic gain by making a specific substitution for the Receiver predictor $b = 2b_U$. Define $Z = b_U^\top X$, so $Z \sim \mathcal{N}(0, \sigma_Z^2)$ and $b^\top X = 2Z$. The functions F and G from the general theory become:

$$\begin{aligned} F(X, U) &= (2U - 2Z) \cdot 2Z, \\ G(X, Y) &= (2Y - 2Z) \cdot 2Z. \end{aligned}$$

The strategic gain is $\mathbb{E}[G \cdot \mathbb{I}\{F > 0\}]$.

We condition on $X = x$ and use the Gaussian conditional distribution $U | X = x \sim \mathcal{N}(\mu_{U|x}, s^2)$ where $\mu_{U|x} = b_U^\top x$. Similarly, the linear projection of Y onto U conditional on X gives:

$$Y = \mu_{Y|x} + \frac{\kappa}{s^2}(U - \mu_{U|x}) + \epsilon,$$

where $\mu_{Y|x} = b_Y^\top x$, and ϵ is independent of U given X .

Define $t(x) = b_U^\top x$ and $M(x) = 2Z = 2b_U^\top x$. The condition $F > 0$ reduces to $M(x)(U - t(x)) > 0$.

We evaluate the conditional expectation $\mathcal{G}(x) = \mathbb{E}[G \mathbb{I}\{F > 0\} | X = x]$ using standard truncated Gaussian identities. For a variable $V \sim \mathcal{N}(\mu, \sigma^2)$ and threshold c , the relevant moments are $\mathbb{E}[\mathbb{I}\{V \geq c\}] = \Phi(\frac{\mu-c}{\sigma})$ and $\mathbb{E}[(V - \mu) \mathbb{I}\{V \geq c\}] = \sigma \phi(\frac{\mu-c}{\sigma})$. Using $\alpha = (\mu_{U|x} - t(x))/s = 0$, the conditional contribution evaluates to:

$$\mathcal{G}(x) = \begin{cases} 2M(x) \left[(\mu_{Y|x} - t(x)) \Phi(\alpha) + \frac{\kappa}{s} \phi(\alpha) \right] & \text{if } M(x) \geq 0, \\ 2M(x) \left[(\mu_{Y|x} - t(x)) (1 - \Phi(\alpha)) - \frac{\kappa}{s} \phi(\alpha) \right] & \text{if } M(x) < 0. \end{cases}$$

To compute the unconditional expectation $\mathbb{E}[\mathcal{G}(X)]$, we separate the quadratic and linear components. For the quadratic term, define $Q(x) = 2M(x)(\mu_{Y|x} - t(x))$. Since $M(x) = 2b_U^\top x$ and $\mu_{Y|x} - t(x) = (b_Y - b_U)^\top x$:

$$Q(x) = 4(b_U^\top x)((b_Y - b_U)^\top x).$$

The contribution from the quadratic term integrates to:

$$\begin{aligned} & \mathbb{E}[Q(X)\Phi(\alpha)\mathbb{I}\{M \geq 0\}] \\ & + \mathbb{E}[Q(X)(1-\Phi(\alpha))\mathbb{I}\{M < 0\}] \\ & = \frac{1}{2}\mathbb{E}[Q(X)] \\ & = 2(b_Y^\top \Sigma_{XU} - \sigma_Z^2), \end{aligned}$$

since $Q(x)$ is symmetric and $\alpha = 0$ gives $\Phi(0) = 1/2$.

For the linear term, we integrate the components with $\phi(\alpha)$:

$$\begin{aligned} & \frac{2\kappa}{s}\phi(\alpha)\left(\mathbb{E}[M(X)\mathbb{I}\{M \geq 0\}] \right. \\ & \quad \left. - \mathbb{E}[M(X)\mathbb{I}\{M < 0\}]\right) \\ & = \frac{4\kappa}{s}\phi(\alpha)\mathbb{E}[2Z\mathbb{I}\{Z \geq 0\}]. \end{aligned}$$

Since $Z \sim \mathcal{N}(0, \sigma_Z^2)$, we have $\mathbb{E}[Z\mathbb{I}\{Z \geq 0\}] = \sigma_Z/\sqrt{2\pi}$. Substituting $\phi(0) = 1/\sqrt{2\pi}$:

$$\text{Linear term} = \frac{8\kappa\sigma_Z}{s\sqrt{2\pi}}e^{-\alpha^2/2}.$$

Combining both terms yields $2(b_Y^\top \Sigma_{XU} - \sigma_Z^2) + \frac{8\kappa\sigma_Z}{s\sqrt{2\pi}}e^{-\alpha^2/2}$. A strategic advantage exists when

$$2(b_Y^\top \Sigma_{XU} - \sigma_Z^2) + \frac{8\kappa\sigma_Z}{s\sqrt{2\pi}}e^{-\alpha^2/2} > \Sigma_{XY}^\top \Sigma_{XX}^{-1} \Sigma_{XY},$$

which is precisely (12). □

6.2 Proofs for Strategic Advantage Condition for Classification

Proof of Lemma 4. Under the mandatory disclosure setup (vanilla benchmark), the Receiver implements $\hat{y}_V(\emptyset) = \text{sign}(\mathbb{E}[Y])$ when $M = \emptyset$, and $\hat{y}_V(x) = \text{sign}(\mathbb{E}[Y | X = x])$ when $M = X$. Let $\eta(x) = \mathbb{E}[Y | X = x]$ and $p(x) = \mathbb{P}(Y = 1 | X = x)$. Since $Y \in \{-1, +1\}$, we have $\eta(x) = 2p(x) - 1$. When $\eta(x) \geq 0$, the optimum prediction is $+1$, yielding an error probability of $1 - p(x)$. When $\eta(x) < 0$, the optimum prediction is -1 , yielding an error probability of $p(x)$. In both cases, the conditional Bayes risk is equivalently given by $(1 - |\eta(x)|)/2$. Thus, the overall expected risk when $M = X$ is $\mathbb{E}[(1 - |\eta(X)|)/2]$.

When $M = \emptyset$, the constant classifier yields risk $\mathbb{P}(Y \neq \text{sign}(\mathbb{E}[Y]))$, which is equal to $\mathbb{P}(Y = -\text{sign}(\mathbb{E}[Y]))$. Combining these mutually exclusive events with their respective probabilities q and $1 - q$ under the noisy channel yields the stated vanilla risk in (13). □

Proof of Lemma 5. For a chosen default prediction $\hat{y}(M) = i$ when $M = \emptyset$, the Receiver correctly learns from a non-empty message $M = X$ that $U = -i$. Consequently, the optimal conditional strategy becomes $\hat{y}_V(x) = \text{sign}(\mathbb{E}[Y | X = x, U = -i])$. If the Sender has preference $U = i$, withholding features gives a loss of 0 (since the Receiver plays i), while revealing may risk a different

prediction. Thus, the Sender strictly withholds the document, causing $M = \emptyset$. The conditional risk for this group ($U = i$) is simply the risk of the default prediction: $\mathbb{P}(Y = -i \mid U = i)$. If the Sender has preference $U = -i$, withholding yields the worst possible loss for them. Hence, they choose to reveal $A = X$. The message drops with probability q (leading to prediction i and contributing $q\mathbb{P}(Y = -i \mid U = -i)$ to the error), and successfully arrives with probability $1 - q$, where the conditional Bayes risk over the subgroup is $\mathbb{E}[(1 - |\eta_{-i}(X)|)/2]$. Multiplying the expectations by their respective cluster probabilities $\pi_i = \mathbb{P}(U = i)$ exactly formulates the overall strategic risk of $\mathcal{S}_i$ as (14). $\square$

Proof of Theorem 4. The risk functions $R_{\mathcal{S}_1}^*(q)$ and $R_{\mathcal{S}_{-1}}^*(q)$ from Lemma 5 are linear equations in q . Assuming $\mathbb{E}[Y] > 0$, which implies $\mathbb{P}(Y = 1) > \mathbb{P}(Y = -1)$. At maximum channel noise ($q = 1$), the messages are completely dropped ($M = \emptyset$), reducing the Receiver risk to the static default prediction's baseline error. Hence, $R_{\mathcal{S}_1}^*(1) = \mathbb{P}(Y = -1)$, and $R_{\mathcal{S}_{-1}}^*(1) = \mathbb{P}(Y = 1)$. Since $\mathbb{P}(Y = 1) > \mathbb{P}(Y = -1)$, it is clear that $R_{\mathcal{S}_1}^*(1) < R_{\mathcal{S}_{-1}}^*(1)$, making $\mathcal{S}_1$ strictly optimal at $q = 1$. Because both risks are lines as functions of q , if $R_{\mathcal{S}_1}^*(0) \leq R_{\mathcal{S}_{-1}}^*(0)$ (lets call it scenarion A), $\mathcal{S}_1$ is optimal at both extremes and thus dominates everywhere via linear interpolation. Conversely, if $R_{\mathcal{S}_{-1}}^*(0) < R_{\mathcal{S}_1}^*(0)$ (lets call it scenario B), the linear functions cross once inside the interval $q \in (0, 1)$. This intersection naturally defines the transition threshold q^* given by

$$q^* = \frac{R_{\mathcal{S}_1}^*(0) - R_{\mathcal{S}_{-1}}^*(0)}{(R_{\mathcal{S}_{-1}}^*(1) - R_{\mathcal{S}_{-1}}^*(0)) - (R_{\mathcal{S}_1}^*(1) - R_{\mathcal{S}_1}^*(0))}, \quad (30)$$

such that for $q < q^*$, $\mathcal{S}_{-1}$ is optimal, and for $q \geq q^*$, $\mathcal{S}_1$ is optimal. We obtain the strategic advantage condition by verifying when $R_V^*(q) - R_{\mathcal{S}_i}^*(q) > 0$ for the optimal i . Under the domain where $\mathcal{S}_1$ is Stackelberg-optimal (all q in scenario A, or $q \geq q^*$ in scenario B):

$$R_V^*(q) - R_{\mathcal{S}_1}^*(q) > 0.$$

Using $\text{sign}(\mathbb{E}[Y]) = 1$, we decompose the baseline error rate in $R_V^*(q)$:

$$\begin{aligned} q\mathbb{P}(Y = -1) &= q[\pi_1\mathbb{P}(Y = -1 \mid U = 1) \\ &\quad + \pi_{-1}\mathbb{P}(Y = -1 \mid U = -1)]. \end{aligned}$$

Upon substituting $R_V^*(q)$ and $R_{\mathcal{S}_1}^*(q)$, the terms matching $q\pi_{-1}\mathbb{P}(Y = -1 \mid U = -1)$ cancel algebraically. The remaining terms structurally share a common $(1 - q)$ factor. Dividing the strict inequality by $(1 - q)$ yields the noise-invariant condition shown in (15). In the alternate domain where $\mathcal{S}_{-1}$ is strictly optimal ($q < q^*$ in scenario B), we calculate $R_V^*(q) - R_{\mathcal{S}_{-1}}^*(q) > 0$. Here, the q dependencies no longer cancel out purely because the vanilla baseline (+1) and the strategic default choice (-1) are misaligned. Grouping the independent expectations corresponding to q and $(1 - q)$ distinctly and isolating terms directly formulates the threshold specified in (17). $\square$

6.3 Static Learning Generalization Bounds

We first state some standard results in learning theory that we will use in our analysis.

Lemma 8. *Let $\mathcal{F} \subseteq \{f : \mathbb{R}^d \rightarrow \mathbb{R}\}$ and $\mathcal{G} \subseteq \{g : \mathbb{R}^d \rightarrow \mathbb{R}\}$ be function classes, and let $\mathcal{Z} = \{z_1, \dots, z_n\}$ be a collection of n points in $\mathbb{R}^d$. Let $d_{VC}(\cdot)$ denote the VC dimension and let $\delta \in (0, 1)$.*

(i) *For all $f \in \mathcal{F}$, with probability at least $1 - \delta$, we have*

$$\mathbb{E}[f(z)] \leq \frac{1}{n} \sum_{\mathcal{Z}} f(z_i) + 2\mathcal{R}(\mathcal{F}; \mathcal{Z}) + 4C \sqrt{\frac{2 \log(4/\delta)}{n}}.$$

- (ii) If $\mathcal{F} = \{\phi \circ g : g \in \mathcal{G}\}$ where ϕ is L -Lipschitz, then $\mathcal{R}(\mathcal{F}; \mathcal{Z}) \leq L\mathcal{R}(\mathcal{G}; \mathcal{Z})$.
- (iii) If $\sup_{x, f \in \mathcal{F}} |f(x)| \leq C_1$ and $\sup_{x, g \in \mathcal{G}} |g(x)| \leq C_2$, then $\mathcal{R}(\mathcal{F} \cdot \mathcal{G}; \mathcal{Z}) \leq C_2 \mathcal{R}(\mathcal{F}; \mathcal{Z}) + C_1 \mathcal{R}(\mathcal{G}; \mathcal{Z})$.
- (iv) The Rademacher complexity and VC dimension satisfy

$$\mathcal{R}(\mathcal{F}; \mathcal{Z}) \leq \mathcal{O}\left(\sqrt{\frac{d_{VC}(\mathcal{F}) \log n}{n}}\right).$$

- (v) Suppose the function class $\mathcal{F} := \{f : f(x) = \mathbb{I}\{Q(x) \geq 0\}, Q(x) = x^\top Ax + b^\top x + c, \text{ where } A \in \mathbb{R}^{d \times d}, b \in \mathbb{R}^d, \text{ and } c \in \mathbb{R}\}$. Then $d_{VC}(\mathcal{F}) \leq \frac{(d+1)(d+2)}{2}$, where d is the dimension of the feature space.
- (vi) Let $\mathcal{F} := \{f : f(x) = a + b^\top x, x \in [-C, C], a \in [-C, C], b \in [-C, C]^d\}$ be a linear function class. Then $\mathcal{R}(\mathcal{F}; \mathcal{Z}) = \frac{dC^2}{\sqrt{n}}$.
- (vii) Let Z be a random variable with $\mathbb{E}[Z] = 0$ and $|Z| \leq C$ a.s. Then with probability at least $1 - \delta$,

$$\left| \frac{1}{n} \sum_{i=1}^n z_i^2 - \text{Var}(Z) \right| \leq C^2 \sqrt{\frac{\log(2/\delta)}{n}}.$$

- (viii) Let $\hat{f} \in \arg \min_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^n f(z_i)$ and $f^* \in \arg \min_{f \in \mathcal{F}} \mathbb{E}[f(z)]$. Then with probability at least $1 - \delta$

$$\mathbb{E}[\hat{f}(z)] \leq \mathbb{E}[f^*(z)] + 2\mathcal{R}(\mathcal{F}; \mathcal{Z}) + \mathcal{O}\left(C \sqrt{\frac{\log(1/\delta)}{n}}\right).$$

Proof. We refer the reader to the standard literature for the detailed proofs of the established claims. Claims (i) and (ii) follow directly from Theorem 26.5(ii) (p. 378) and Lemma 26.9 (p. 381) in Chapter 26 of [?], respectively. Claim (iii) follows by applying the contraction Lemma (Claim (ii)) as follows. For any $f \in \mathcal{F}$ bounded by C_1 and $g \in \mathcal{G}$ bounded by C_2 , the product mapping $\phi(x, y) = x \cdot y$ is Lipschitz. By fixing one function and bounding the variation of the other, the Lipschitz constant with respect to f is bounded by $\sup |g| \leq C_2$, and with respect to g is bounded by $\sup |f| \leq C_1$. Applying the contraction property yields $\mathcal{R}(\mathcal{F} \cdot \mathcal{G}; \mathcal{Z}) \leq C_2 \mathcal{R}(\mathcal{F}; \mathcal{Z}) + C_1 \mathcal{R}(\mathcal{G}; \mathcal{Z})$. Claim (iv) is derived by combining Sauer's Lemma and Massart's Lemma. Claim (v) is established using the polynomial embedding trick. A quadratic function $Q(x) = x^\top Ax + b^\top x + c$ in $\mathbb{R}^d$ can be expressed as a linear inner product $\langle \tilde{w}, \tilde{x} \rangle$ in a lifted feature space. The lifted vector $\tilde{x}$ contains d linear terms x_i , $\frac{d(d+1)}{2}$ quadratic cross-terms $x_i x_j$ (for $i \leq j$), and 1 constant term. The dimension of this new space is

$$D = \frac{d(d+1)}{2} + d + 1 = \frac{d^2 + 3d + 2}{2} = \frac{(d+1)(d+2)}{2}.$$

Since linear halfspaces in $\mathbb{R}^D$ have a VC dimension equal to D , the bound on $d_{VC}(\mathcal{F})$ follows. Claim (vi) follows directly from Theorem 26.10 (p. 382) in [?]. Claim (vii) is a standard consequence of Hoeffding's concentration inequality. Since $|Z| \leq C$ almost surely, the squared random variable Z^2 is bounded in the interval $[0, C^2]$. Applying Hoeffding's inequality to the empirical mean of Z^2 gives

$$\mathbb{P}\left(\left|\frac{1}{n} \sum_{i=1}^n z_i^2 - \mathbb{E}[Z^2]\right| > \epsilon\right) \leq 2 \exp\left(-\frac{2n\epsilon^2}{C^4}\right).$$

Setting the right-hand side equal to δ and solving for ϵ yields the stated bound. Finally, Claim (viii) follows directly from Theorem 26.5(iii) (p. 378) in [?]. $\square$

Proof of Lemma 6. Let $\hat{L} = \frac{1}{n} \sum_{\mathcal{S}} (y_i - b^\top x_i)^2$ and $L = \mathbb{E}[(Y - b^\top X)^2]$. We can rewrite the generalization gap as

$$R_V(b, q) - \hat{R}_V(b, q) \leq |\text{Var}(Y) - \frac{1}{n} \sum_{\mathcal{S}} y_i^2| + (1 - q)(L - \hat{L}).$$

Applying a union bound, we allocate a failure probability of $\delta/2$ to each term. From Lemma 8 (vii), the variance estimation error is bounded by $\mathcal{O}\left(C^2 \sqrt{\log(1/\delta)/n}\right)$.

We now bound $L - \hat{L}$. Let $\mathcal{H} := \{h_b(x, y) : h_b(x, y) = y - b^\top x, x \in \mathbb{R}^d, b \in \mathbb{R}^d, y \in \mathbb{R}\}$ be a linear function class. Recall that $\ell(a, b) = (a - b)^2$ is a quadratic loss function. Because $Y \in [-C, C]$ and $b^\top X \in [-dC^2, dC^2]$, the maximum derivative of $\ell(y, b^\top x)$ is $2(dC^2 + C) = 2C(dC + 1)$. Thus, ℓ is $2C(dC + 1)$ -Lipschitz. By Lemma 8 (ii) and (vi), we have $\mathcal{R}(\ell \circ \mathcal{H}; \mathcal{Z}) \leq 2C(dC + 1) \frac{dC^2}{\sqrt{n}} = \mathcal{O}\left(\frac{d^2 C^4}{\sqrt{n}}\right)$.

Applying the Rademacher generalization bound from Lemma 8 (i), the term $L - \hat{L}$ is bounded by $2\mathcal{R}(\ell \circ \mathcal{H}; \mathcal{Z}) + \mathcal{O}\left(C^2 \sqrt{\log(1/\delta)/n}\right)$. Combining the variance and loss terms yields the desired $\mathcal{O}\left(\frac{d^2 C^4}{\sqrt{n}} + C^2 \sqrt{\frac{\log(1/\delta)}{n}}\right)$ bound. $\square$

Proof of Lemma 7. Applying the triangle inequality, the difference $R_S(b, q) - \hat{R}_S(b, q)$ can be upper bounded as

$$\begin{aligned} R_S(b, q) - \hat{R}_S(b, q) &\leq |\text{Var}(Y) - \frac{1}{n} \sum_{i \in \mathcal{S}} y_i^2| + (1 - q) \\ &\quad \times \left| \mathbb{E}[G(X, Y) \mathbb{I}\{F(U, X) > 0\}] \right. \\ &\quad \left. - \frac{1}{n} \sum_{i \in \mathcal{S}} \mathbb{I}\{F(u_i, x_i) > 0\} G(x_i, y_i) \right|. \end{aligned}$$

Applying a union bound, we allocate a failure probability of $\delta/2$ to the variance term and $\delta/2$ to the expectation term. The variance term is bounded by $\mathcal{O}\left(C^2 \sqrt{\log(1/\delta)/n}\right)$ from Lemma 8 (vii).

To bound the expectation term uniformly, we define the hypothesis classes $\mathcal{F} := \{f_b : f_b(u, x) = \mathbb{I}\{(2u - b^\top x)b^\top x > 0\}, b \in \mathbb{R}^d\}$ and $\mathcal{G} := \{g_b : g_b(x, y) = (2y - b^\top x)b^\top x, b \in \mathbb{R}^d\}$. Utilizing the Rademacher generalization bound from Lemma 8 (i), the gap is bounded by $2\mathcal{R}(\mathcal{F} \cdot \mathcal{G}; \mathcal{Z}) + \mathcal{O}\left(C^2 \sqrt{\log(1/\delta)/n}\right)$.

To bound the Rademacher complexity of the product class $\mathcal{F} \cdot \mathcal{G}$, we break down the calculation and establish individual bounds for $\mathcal{R}(\mathcal{F}; \mathcal{Z})$ and $\mathcal{R}(\mathcal{G}; \mathcal{Z})$ in the following claims.

Claim 1. $\mathcal{R}(\mathcal{F}; \mathcal{Z}) \leq \mathcal{O}\left(d \sqrt{\frac{\log n}{n}}\right)$.

Proof of Claim 1. From Lemma 8 (iv), $\mathcal{R}(\mathcal{F}; \mathcal{Z}) \leq \mathcal{O}\left(\sqrt{\frac{d_{\text{VC}}(\mathcal{F}) \log n}{n}}\right)$. Because the decision boundary of $\mathcal{F}$ is quadratic in the feature space $\mathbb{R}^{d+1}$, from Lemma 8 (v), $d_{\text{VC}} \leq \frac{(d+2)(d+3)}{2} = \mathcal{O}(d^2)$. Substituting this quadratic bound into the square root yields $\sqrt{\mathcal{O}(d^2) \log(n)/n}$, which simplifies to $\mathcal{O}\left(d \sqrt{\log n/n}\right)$. $\square$

Claim 2. $\mathcal{R}(\mathcal{G}; \mathcal{Z}) \leq \mathcal{O}\left(\frac{d^2 C^4}{\sqrt{n}}\right)$.

Proof of Claim 2. Let $\phi_i(z) := (2y_i - z)z$ and $\mathcal{H} := \{h_b : h_b(x) = b^\top x, b \in \mathbb{R}^d\}$. From the definition of $\mathcal{G}$, $g_b(x_i, y_i) = \phi_i(h_b(x_i))$. Since $|Y| \leq C$ and $|Z| = |b^\top X| \leq dC^2$, the maximum derivative of $\phi_i(z)$

is bounded by $2C + 2dC^2 \leq 2C(1 + dC)$. Thus, the function $\phi_i(z)$ is $2C(1 + dC)$ -Lipschitz. Applying Lemma 8 (ii) yields $\mathcal{R}(\mathcal{G}; \mathcal{Z}) \leq 2C(1 + dC)\mathcal{R}(\mathcal{H}; \mathcal{Z})$. Since $\mathcal{H}$ is a linear class, $\mathcal{R}(\mathcal{H}; \mathcal{Z}) \leq \frac{dC^2}{\sqrt{n}}$, and the claim holds. $\square$

Because functions in $\mathcal{F}$ are bounded by 1, and functions in $\mathcal{G}$ are bounded by $\mathcal{O}(d^2C^4)$, from Lemma 8 (iii), we get $\mathcal{R}(\mathcal{F} \cdot \mathcal{G}; \mathcal{Z}) \leq \mathcal{O}(d^2C^4)\mathcal{R}(\mathcal{F}; \mathcal{Z}) + 1 \cdot \mathcal{R}(\mathcal{G}; \mathcal{Z})$. Substituting the individual bounds yields $\mathcal{O}\left(d^3C^4\sqrt{\frac{\log n}{n}}\right)$. Combining the Rademacher complexity with the concentration terms completes the proof. $\square$

Proof of Theorem 5. The absolute difference $|\Delta^* - \widehat{\Delta}(\mathcal{S})|$ is upper bounded by the triangle inequality

$$|R_V^*(q) - \hat{R}_V(\hat{b}_V, q)| + |R_S^*(q) - \hat{R}_S(\hat{b}_S, q)|.$$

We upper bound each of these terms using Lemma 8 (viii). Note that the right-hand side of this lemma is of the same order as Lemma 8 (i). Thus we apply the bounds derived in Lemma 6 for the vanilla risk term and Lemma 7 for the strategic risk term. We then take a union bound of $\delta/2$ for each term and absorb the lower-order vanilla risk term into the strategic risk term to arrive at the final bound. $\square$

Proof of Theorem 6. Let $\epsilon(n, \delta) = \mathcal{O}\left(d^3C^4\sqrt{\frac{\log n}{n}} + C^2\sqrt{\frac{\log(1/\delta)}{n}}\right)$. By a union bound over the preceding lemmas, the generalization errors for both disclosure regimes and the advantage estimation error $|\Delta^* - \widehat{\Delta}(\mathcal{S})|$ are simultaneously bounded by $\epsilon(n, \delta)$ with probability at least $1 - \delta$.

If the empirical advantage satisfies $|\widehat{\Delta}(\mathcal{S})| > \epsilon(n, \delta)$, the estimation error is insufficient to flip the sign of the advantage. Consequently, the algorithm identifies the optimal disclosure regime with certainty. In this scenario, the total excess risk is bounded solely by the generalization gap of the chosen regime, which is at most $\epsilon(n, \delta)$.

Otherwise, if $|\widehat{\Delta}(\mathcal{S})| \leq \epsilon(n, \delta)$, the algorithm may select the suboptimal disclosure regime. This selection incurs an additional penalty equal to the true strategic advantage $|\Delta^*|$. Since $|\Delta^*| \leq |\widehat{\Delta}(\mathcal{S})| + \epsilon(n, \delta) \leq 2\epsilon(n, \delta)$, the total excess risk is bounded by the sum of the generalization error and the selection penalty, totaling $3\epsilon(n, \delta)$.

In both scenarios, the excess risk is $\mathcal{O}(\epsilon(n, \delta))$. Absorbing the constants yields the final bound as claimed. $\square$

6.4 Online Learning Analysis

Proof of Theorem 7 (Full Feedback Convergence). At each round t , the receiver performs the update: $b_{t+1} = \Pi_{\mathcal{B}}(b_t - \eta_t g_t)$, where g_t is an unbiased estimate of $\nabla \mathcal{L}(b_t)$. We analyze the distance to the optimum b^* :

$$\begin{aligned} \|b_{t+1} - b^*\|^2 &= \|\Pi_{\mathcal{B}}(b_t - \eta_t g_t) - b^*\|^2 \leq \|b_t - \eta_t g_t - b^*\|^2 \\ &= \|b_t - b^*\|^2 - 2\eta_t g_t^\top (b_t - b^*) + \eta_t^2 \|g_t\|^2. \end{aligned}$$

Rearranging terms and taking the expectation over the randomness at round t :

$$\mathbb{E}[\nabla \mathcal{L}(b_t)^\top (b_t - b^*)] \leq \frac{\mathbb{E}[\|b_t - b^*\|^2] - \mathbb{E}[\|b_{t+1} - b^*\|^2]}{2\eta_t} + \frac{\eta_t G^2}{2},$$

where we bounded the gradients as $\|g_t\| \leq G$. Given our assumption that b_t remains within the locally convex basin, we apply the first-order convexity condition $\mathcal{L}(b_t) - \mathcal{L}(b^*) \leq \nabla \mathcal{L}(b_t)^\top (b_t - b^*)$. Summing

from $t = 1$ to T with a fixed learning rate $\eta_t = \eta = \frac{D}{G\sqrt{T}}$ and bounded domain $\|b_1 - b^*\| \leq D$:

$$\frac{1}{T} \sum_{t=1}^T \mathbb{E}[\mathcal{L}(b_t) - \mathcal{L}(b^*)] \leq \frac{D^2}{2\eta T} + \frac{\eta G^2}{2} = \frac{DG}{\sqrt{T}}.$$

This confirms that with full information, the receiver learns the optimal strategy at an $\mathcal{O}(1/\sqrt{T})$ rate. $\square$

Proof of Theorem 8 (Partial Feedback Convergence). To evaluate the convergence to the true local minimum b^* , we decompose the excess risk using the triangle inequality:

$$\begin{aligned} \mathbb{E}[\mathcal{L}(b_t) - \mathcal{L}(b^*)] &\leq \mathbb{E}[|\mathcal{L}(b_t) - \tilde{\mathcal{L}}(b_t)|] \\ &\quad + \mathbb{E}[\tilde{\mathcal{L}}(b_t) - \tilde{\mathcal{L}}(\tilde{b}^*)] \\ &\quad + |\tilde{\mathcal{L}}(\tilde{b}^*) - \mathcal{L}(b^*)|. \end{aligned}$$

We evaluate each term as follows:

- **Landscape Bias:** Since $|U - Y| \leq \epsilon$ in L^2 -norm, and given K -Lipschitzness of the risk in its distribution parameters:

$$\mathbb{E}[|\mathcal{L}(b_t) - \tilde{\mathcal{L}}(b_t)|] \leq K\|U - Y\|_{L^2} \leq K\epsilon.$$

- **Surrogate Regret:** Bound by $\mathcal{O}(1/(p_{\min}\sqrt{T}))$ from the standard OGD guarantee under partial observation (similar to Theorem 7, scaled by the minimum revelation probability $p_{\min}$).
- **Minima Gap:** We expand the gap between the values at local optima using the triangle inequality:

$$|\tilde{\mathcal{L}}(\tilde{b}^*) - \mathcal{L}(b^*)| \leq |\tilde{\mathcal{L}}(\tilde{b}^*) - \tilde{\mathcal{L}}(b^*)| + |\tilde{\mathcal{L}}(b^*) - \mathcal{L}(b^*)|.$$

By uniform bounds on the difference between true and surrogate risk functions:

$$|\tilde{\mathcal{L}}(b^*) - \mathcal{L}(b^*)| \leq K\|U - Y\|_{L^2} \leq K\epsilon.$$

To evaluate $|\tilde{\mathcal{L}}(\tilde{b}^*) - \tilde{\mathcal{L}}(b^*)|$, we first bound the distance between the two minima $\|b^* - \tilde{b}^*\|_2$. Assuming $\mathcal{L}(b)$ is locally λ -strongly convex, we have:

$$(\nabla \mathcal{L}(b^*) - \nabla \mathcal{L}(\tilde{b}^*))^\top (b^* - \tilde{b}^*) \geq \lambda \|b^* - \tilde{b}^*\|_2^2.$$

By the Cauchy-Schwarz inequality, this implies $\|\nabla \mathcal{L}(b^*) - \nabla \mathcal{L}(\tilde{b}^*)\|_2 \geq \lambda \|b^* - \tilde{b}^*\|_2$. Since b^* is a local minimum, $\nabla \mathcal{L}(b^*) = 0$, giving $\|\nabla \mathcal{L}(\tilde{b}^*)\|_2 \geq \lambda \|b^* - \tilde{b}^*\|_2$.

Similarly, since $\tilde{b}^*$ is a local minimum of the surrogate risk, $\nabla \tilde{\mathcal{L}}(\tilde{b}^*) = 0$. Assuming the gradient perturbation is bounded by $K\epsilon$, we get:

$$\|\nabla \mathcal{L}(\tilde{b}^*)\|_2 = \|\nabla \mathcal{L}(\tilde{b}^*) - \nabla \tilde{\mathcal{L}}(\tilde{b}^*)\|_2 \leq K\epsilon.$$

Combining these yields the distance between the local minima:

$$\|b^* - \tilde{b}^*\|_2 \leq \frac{K\epsilon}{\lambda}.$$

Next, we utilize the L -smoothness of the surrogate risk $\tilde{\mathcal{L}}$ around its minimum $\tilde{b}^*$:

$$\tilde{\mathcal{L}}(b^*) - \tilde{\mathcal{L}}(\tilde{b}^*) \leq \nabla \tilde{\mathcal{L}}(\tilde{b}^*)^\top (b^* - \tilde{b}^*) + \frac{L}{2} \|b^* - \tilde{b}^*\|_2^2.$$

With $\nabla \tilde{\mathcal{L}}(\tilde{b}^*) = 0$, the optimality difference is bounded by:

$$|\tilde{\mathcal{L}}(\tilde{b}^*) - \tilde{\mathcal{L}}(b^*)| \leq \frac{L}{2} \|b^* - \tilde{b}^*\|_2^2 \leq \frac{LK^2\epsilon^2}{2\lambda^2}.$$

Therefore, the total minima gap is bounded by $\frac{LK^2\epsilon^2}{2\lambda^2} + K\epsilon$.

Summing the terms, the total expected average excess strategic risk is bounded by:

$$\frac{1}{T} \sum_{t=1}^T \mathbb{E}[\mathcal{L}(b_t) - \mathcal{L}(b^*)] \leq \mathcal{O}\left(\frac{1}{\sqrt{T}}\right) + \max\{\mathcal{O}(\epsilon^2), \mathcal{O}(\epsilon)\}.$$

This completes the proof. □

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